Web Technology Study Notes

Module I: Introduction to Web and Internet Fundamentals

1. Evolution of the Web

ARPANET (Advanced Research Projects Agency Network): Developed by the US Department of Defense in the late 1960s, ARPANET was the foundational network that pioneered packet switching technology. It laid the technological groundwork for today's global Internet.

Web 1.0 (The Read-Only Web): Spanning roughly from 1991 to 2004, Web 1.0 consisted primarily of static websites built with HTML. Content creators were limited, and the vast majority of users were passive consumers of information with no user-to-user interactivity.

Web 2.0 (The Social/Read-Write Web): Popularized in the mid-2000s, Web 2.0 shifted focus toward user-generated content, usability, and interoperability. Technologies like JavaScript, AJAX, XML, and CSS enabled dynamic content, paving the way for social media platforms, blogs, wikis, and interactive web apps.

Web 3.0 (The Executable/Semantic/Decentralized Web): Web 3.0 introduces a smarter web driven by structured data representation (Semantic Web) and decentralized networks (Blockchain). It aims to make web data machine-readable while giving users true ownership of data, identities, and digital assets.

Web 4.0 (The Intelligent/Symbiotic Web): An ultra-connected, symbiotic operating system where humans and machines interact in perfect synergy. Characterized by mass AI integration, ubiquitous mobile connectivity, advanced automation, real-time context-awareness, and smart environments.

AI in Web Architecture: Modern web architectures heavily embed Artificial Intelligence to enhance performance and user experience. Examples include automated load balancing, predictive resource caching, conversational AI bots, tailored recommendation engines, and edge-computing AI security filters.

2. Client-Server Model & Communications

Client-Server Model: The foundational distributed application architecture dividing tasks between resource providers (**servers**) and service requesters (**clients**/browsers). The client initiates requests, and the server processes them to provide responses.

Browser-Server Communication Steps:

- **1. URL Processing:** The user enters a URL into the browser window.
- **2. DNS Resolution:** The browser requests the corresponding numeric IP address from a DNS server.
- **3. TCP Handshake:** A persistent connection is established via TCP/IP protocol.
- **4. HTTP Request:** The browser transmits an HTTP/HTTPS request to the destination server.
- **5. Server Processing:** The web server processes the request, communicating with databases if needed.
- **6. HTTP Response:** The server sends back an HTTP response containing headers and data payload.
- **7. Rendering:** The browser parses HTML, CSS, and JS files to render the visual webpage.

DNS (Domain Name System): Known as the 'phonebook of the Internet', DNS translates human-readable domain names (e.g., www.example.com) into machine-readable IP addresses (e.g., 192.0.2.1) so browsers can

route network traffic properly.

3. Web Protocols & State Management

HTTP vs. HTTPS: Hypertext Transfer Protocol (HTTP) handles data exchange in plaintext, leaving it vulnerable to interception. HTTPS (HTTP Secure) encrypts all data transmissions using TLS/SSL cryptographic protocols, establishing identity authentication and message integrity.

Request-Response Cycle: A standardized sequence where a client application transmits a request envelope consisting of a Method (GET, POST, PUT, DELETE), Headers, and optional Body data. The server returns a status package (e.g., 200 OK, 404 Not Found, 500 Server Error) accompanied by response headers and resources.

Cookies vs. Sessions: Since HTTP is inherently stateless, cookies and sessions track user states:

Feature	Cookies	Sessions
Storage Location	Client-side (User Browser)	Server-side
Security	Lower (Vulnerable to tampering)	Higher (Client only holds Session ID)
Data Capacity	Limited (Usually up to 4KB per cookie)	Virtually unlimited (Server storage limits)

4. Modern Industry Trends

Cloud-Native Applications: Systems purposefully designed, built, and deployed to operate natively within cloud computing models. Characterized by microservices architectures, containerization (Docker, Kubernetes), continuous integration/deployment (CI/CD), and elastic scaling.

Semantic Web: An extension of the World Wide Web promoted by the W3C that structure page data using frameworks like RDF, OWL, and XML. By embedding machine-interpretable semantics, metadata can be shared across diverse applications, enterprises, and communities.

Blockchain Basics in Web 3.0: Provides the backbone for decentralized web infrastructure. Utilizing immutable distributed ledgers, decentralized cryptographic identification, and autonomous smart contracts, it eliminates central authorities, enabling trustless peer-to-peer applications.

AI responses may include mistakes. This material is synthesized for educational reference.